MSIN0097 Predictive Analytics — Individual Coursework

Predicting Household Purchasing Power from Spending Patterns

Anonymous student number: TYPS3
Word count: 1,967
Github link: https://github.com/Maywarin-T/predictive-analytics-coursework

1. Problem Framing

Economic policy rests on knowing who is poor, yet this knowledge is surprisingly difficult to obtain. In 2023, over 13 million people in the United Kingdom lived in relative poverty, and an estimated 40 per cent of eligible households did not claim means-tested support (DWP, 2024; Policy in Practice, 2022). If income is hard to observe directly, spending is not: every household leaves a fingerprint in its expenditure. Indeed, consumption has been argued to be a more reliable welfare measure than income, as it better reflects permanent resources and smooths transitory shocks (Meyer and Sullivan, 2023; Brewer, Etheridge and O'Dea, 2017). This project exploits that insight, asking whether purchasing power can be inferred from what households spend rather than what they earn.

The analysis draws on the Living Costs and Food Survey (LCFS) for 2021 to 2023, a nationally representative diary survey covering 14,294 households across the post-COVID recovery and the cost-of-living crisis (ONS and DEFRA, 2025). Rather than using raw income, the target variable is equivalised income: gross household income divided by the OECD modified equivalence scale (1.0 for the first adult, +0.5 per additional adult, +0.3 per child under 14), thereby adjusting for household size and composition (OECD, 2013). This reverses conventional economic analysis: instead of predicting spending from income, the model infers income from spending.

The task is formalised as multi-class classification into five equivalised income quintiles (Q1 = lowest purchasing power through Q5 = highest), evaluated by macro F1-score to ensure all quintiles receive equal weight. All income-derived variables are excluded to prevent target leakage. This framing carries direct policy implications. A household that declares Q1 income to HMRC while exhibiting expenditure patterns characteristic of Q5 presents a discrepancy warranting investigation for potential tax non-compliance. More broadly, the ability to infer purchasing power from spending alone enables proactive welfare targeting and real-time cost-of-living monitoring without requiring income disclosure.

Throughout this project, Claude Code was employed as an agent tool to scaffold the code structure, suggest model architectures, and generate baseline pipelines. Every output was verified for leakage, statistical soundness, and domain correctness. The Appendix documents what was accepted, modified, or rejected and why.

2. Exploratory Data Analysis

The merged dataset presents 1,915 columns, a feature space in which target leakage poses a significant risk. Rather than hand-picking features, Pearson correlations were computed between every column and equivalised income. This analysis identified 227 variables that encode income directly or indirectly: the p300–p499 range (income aggregates and tax deductions, with p344p reaching r = 0.999), all b-prefix columns (means-tested benefits whose amounts are determined by income), and several administrative codes. After exclusion, 52 clean features remained: 27 expenditure categories and 25 demographic indicators, each selected with a documented correlation coefficient.

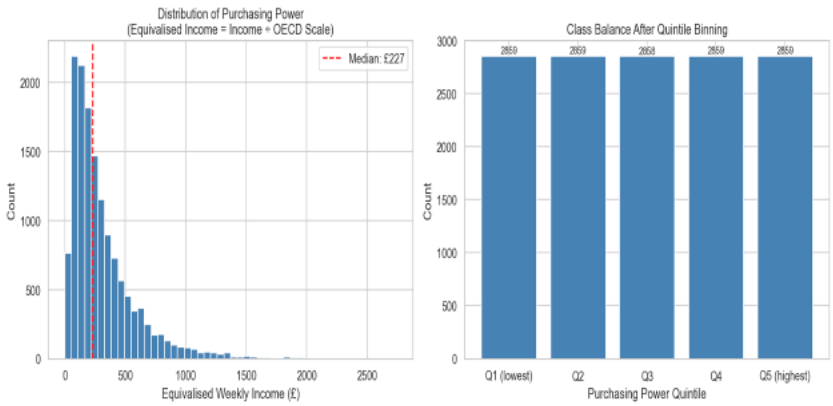

Figure 1: Left panel shows right-skewed equivalised income distribution. Right panel shows balanced quintile classes

Equivalised income has a mean of £311.93 per week and a median of £226.78 per week, exhibiting pronounced right skew (range £0 to £2,740). Quintile binning yields balanced classes of approximately 2,859 households each, thereby avoiding the class imbalance that would arise from fixed income thresholds.

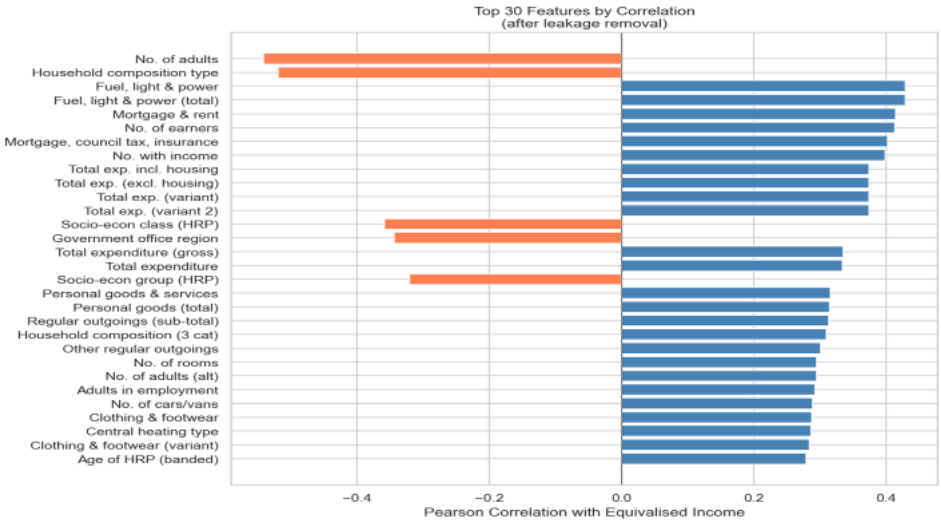

Figure 2: Systematic feature selection — correlation with equivalised income across all non-leakage columns.

Beneath these clean distributions, the data concealed quality problems invisible to standard diagnostics. Standard null-value detection reported zero per cent missing, yet deeper inspection revealed empty strings masquerading as valid entries, 90 negative expenditure values across five columns (likely refunds signalling prior large purchases), and high zero rates in mortgage payments (88.1%), public transport (89.5%), and council tax (64.1%). Importantly, these zeros represent valid economic signals rather than missing data; a zero mortgage payment, for instance, identifies an outright homeowner. Empty strings were converted to NaN for imputation; negatives were retained.

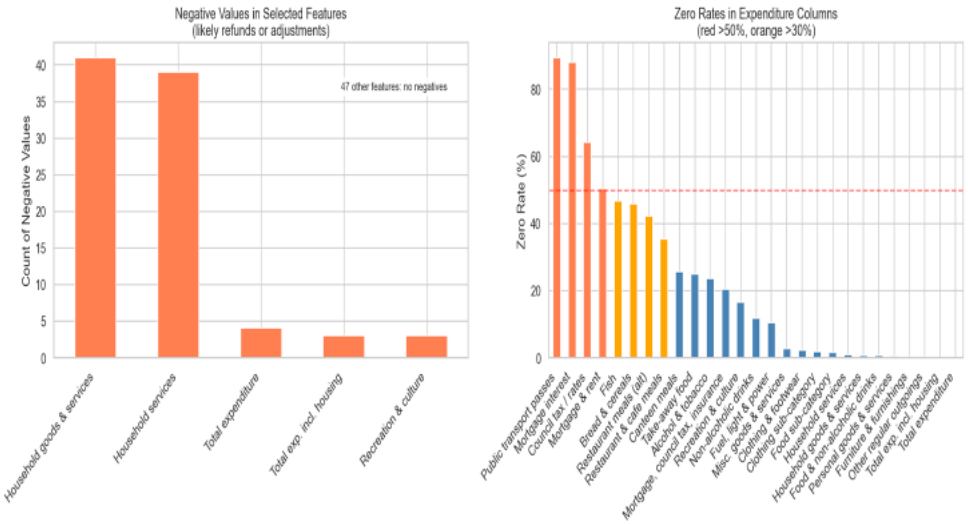

Figure 3: Effective missingness (left) and zero rates (right) revealing hidden data quality issues.

Perhaps the most analytically significant finding concerns how spending differentiates income tiers. Q5/Q1 ratios reveal that mortgage interest varies 200-fold between the richest and poorest quintiles, mortgage and rent 29-fold, and regular outgoings 16-fold. Food, by contrast, differs only 1.8-fold, consistent with Engel's Law: all households require food, yet lower-income households allocate a disproportionately larger share of expenditure to this category (Engel, 1857). These patterns confirm the central premise: spending encodes purchasing power, and the signal is strongest in discretionary categories.

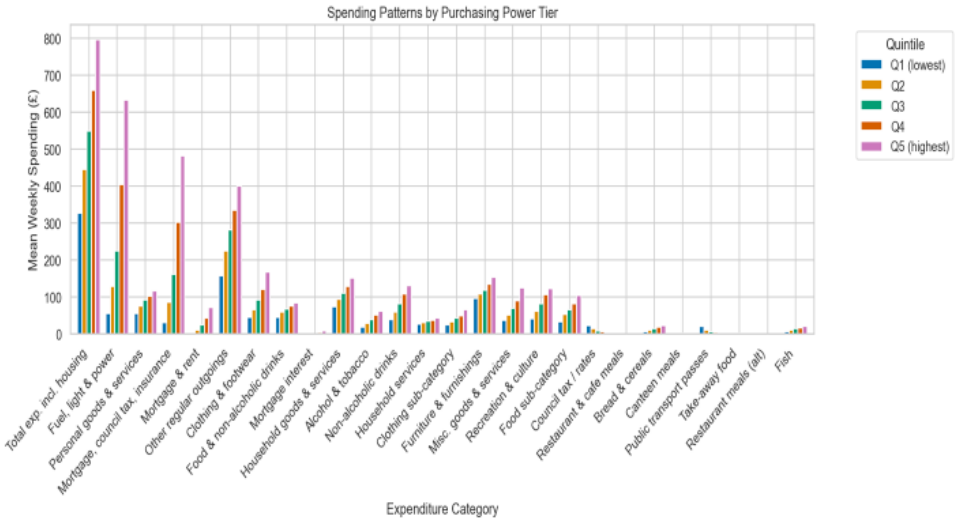

Figure 4: Mean weekly spending by purchasing power tier across expenditure categories.

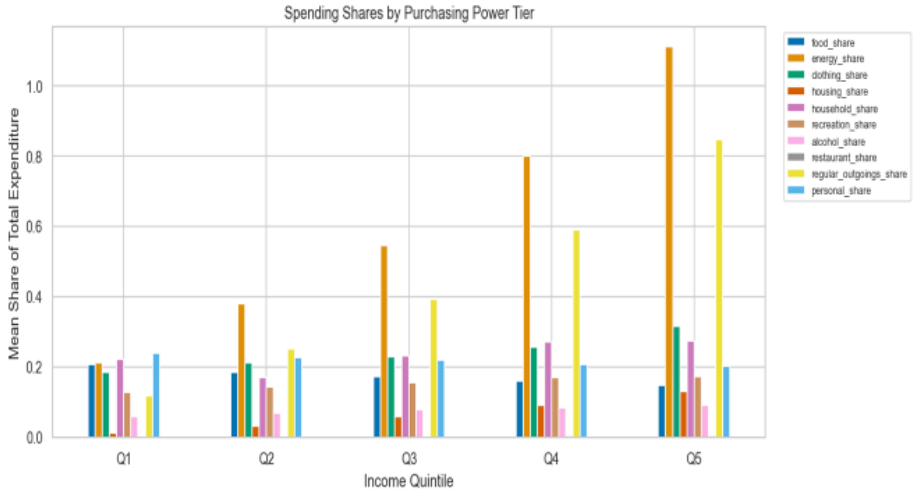

Figure 5: Spending share ratios by purchasing power tier, illustrating Engel's Law pattern.

3. Data Preparation

A temporal split was adopted for this analysis, with 2021–2022 data allocated to training and validation (8,578 + 1,514 households) and the entire 2023 survey reserved for testing (4,202 households). This design forces the model to generalise across spending patterns shaped by the cost-of-living crisis.

The preprocessing pipeline uses scikit-learn's ColumnTransformer (Pedregosa *et al.*, 2011) for reproducibility. For the 38 continuous variables (28 expenditure and demographic measures plus 10 engineered spending-share ratios), a custom CoerceNumeric transformer addressed empty strings that disrupted sklearn's SimpleImputer, followed by median imputation and standard scaling. The 24 categorical variables received mode imputation and one-hot encoding, with unseen categories at test time handled gracefully, producing 204 preprocessed features in total.

Ten spending-share ratios (such as the ratio of food expenditure to total expenditure, energy to total, and housing to total) were additionally engineered, motivated by Engel's Law: lower-income households allocate a higher proportion of spending to necessities, even where absolute expenditure amounts are comparable. Among these ratios, housing share exhibits the strongest Q5/Q1 differential (12.9x), followed by regular outgoings share (7.3x).

To further focus the model on the strongest signals, a preliminary Gradient Boosting classifier ranked all 204 features by importance. The top 35 were retained, led by mortgage, council tax, and insurance (importance = 0.354). This reduction in dimensionality from 204 to 35 features mitigates overfitting while preserving the most informative predictors.

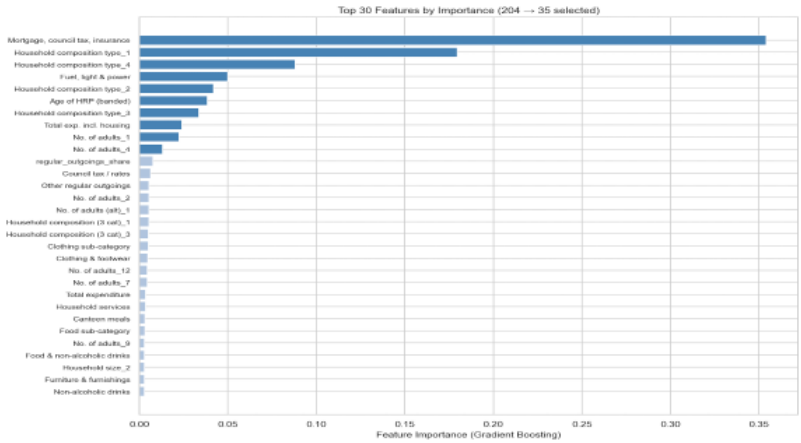

Figure 6: Feature importance ranking used for selection (204 to 35 features).

Regional analysis further contextualises these patterns: London and the South East exhibit the highest purchasing power, while the North East and Wales have the lowest, reinforcing the case for regionally sensitive welfare screening.

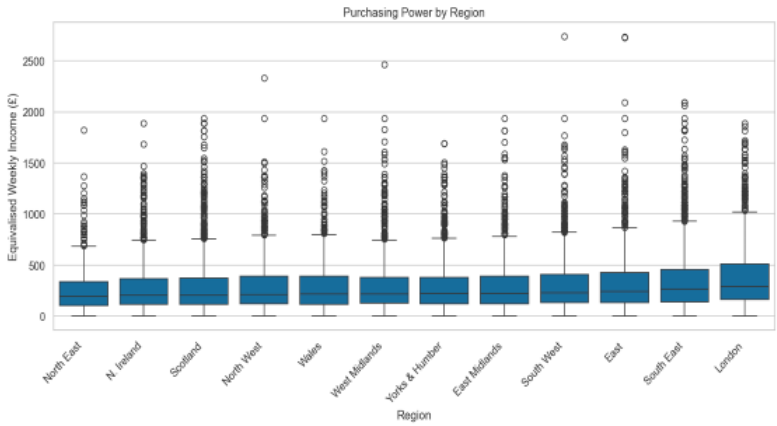

Figure 7: Purchasing power distribution by UK government office region.

To ensure pipeline integrity, eighteen automated pipeline validation tests were implemented, all of which passed. These verify that no NaN values survive preprocessing, no households appear in both training and test sets, class proportions are preserved across splits, and no income-derived leakage variable remains in the feature set.

4. Model Exploration

The exploratory analysis revealed non-linear spending patterns, mixed feature types, and overlapping middle quintiles, characteristics that favour flexible, ensemble-based learners over simple linear models. Six architectures were compared using five-fold stratified cross-validation alongside held-out validation:

Model	CV F1 (±SD)	Val F1	Val Acc	Role
Neural Network	0.7001 ± 0.007	0.7007	0.6988	Deep learning
Gradient Boosting	0.6802 ± 0.005	0.6722	0.6711	Ensemble baseline
SVM (RBF)	0.6713 ± 0.011	0.6762	N/A	Non-linear comparison
Random Forest	0.6642 ± 0.005	0.6594	0.6579	Tree ensemble
Logistic Regression	0.6520 ± 0.014	0.6555	0.6585	Linear baseline
PCA + LogReg	0.3838 ± 0.010	0.3936	0.4108	Dim. reduction

Table 1: Model comparison by cross-validation and validation macro F1.

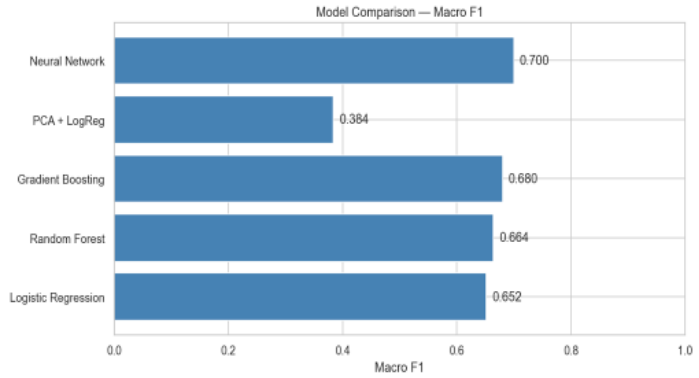

Figure 8: Model comparison by macro F1 score. Neural Network leads all models.

A clear hierarchy emerges. PCA + Logistic Regression collapses to F1 = 0.39, confirming that compressing 38 features into 22 principal components discards the fine-grained expenditure detail needed for quintile separation and that categorical variables, excluded from PCA, carry essential signal. Logistic Regression (F1 = 0.66) provides the lower bound, while the tree ensembles and SVM cluster between 0.66 and 0.68. SVM (Val F1 = 0.68) outperformed both Logistic Regression and Random Forest, lending empirical support to the hypothesis that non-linear models are better suited to this task. However, SVM was not selected for the final pipeline owing to its lack of feature importance measures and native probability outputs.

The Neural Network (64 to 32 units, ReLU, dropout 0.4/0.3) achieved the highest validation performance at F1 = 0.7007 (Chollet *et al.*, 2015). Trained with sparse categorical cross-entropy and early stopping (patience = 10), it manages the bias-variance trade-off more effectively than the tree ensembles (Geron, 2022).

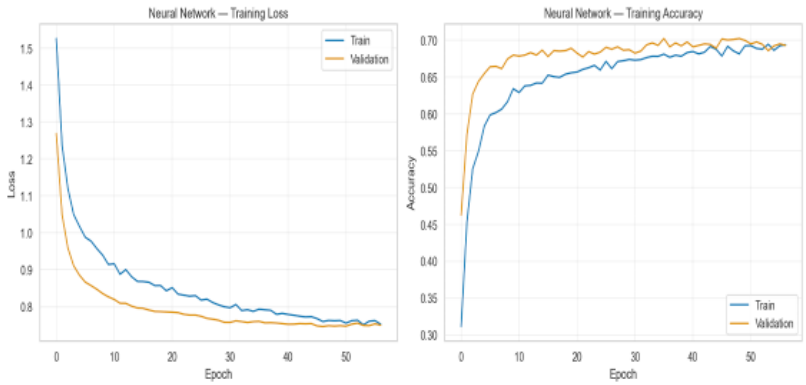

Figure 9: Neural Network training curves — loss (left) and accuracy (right). Early stopping triggers after convergence.

Learning curves for the tree ensembles revealed substantial train-validation gaps, necessitating tighter regularisation, feature selection (204 to 35), and early stopping.

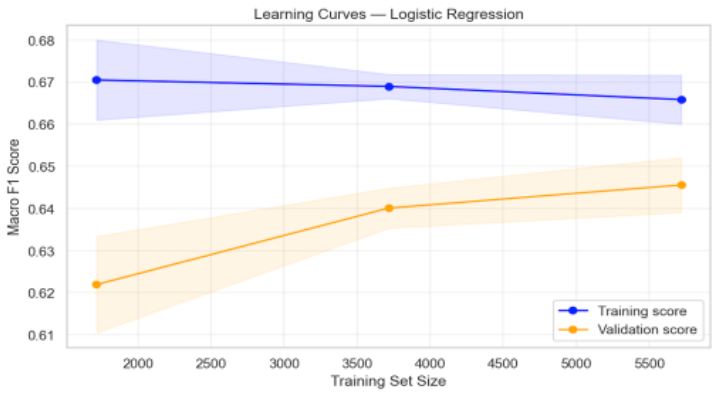

Figure 10: Gradient Boosting learning curves showing train-validation gap.

A notable agent error was identified during hyperparameter tuning, where the agent set the tuning objective to accuracy rather than macro F1. Although the numerical impact is small with balanced classes, the conceptual error is significant, as accuracy disproportionately rewards correct classification of the majority class, whereas macro F1 assigns equal weight to all quintiles. This was identified during code review and corrected before final evaluation.

5. Fine-Tuning and Evaluation

Hyperparameter tuning was applied to all scikit-learn models (Geron, 2022): Logistic Regression via grid search (optimal regularisation strength C = 50.0), Random Forest via randomised search (optimal depth of 20 with balanced subsample weighting), and Gradient Boosting via randomised search (learning rate of 0.05, depth of 4, 500 estimators, 70% subsampling). Tuning improved validation F1 for all three models (Gradient Boosting from 0.6722 to 0.6798, Random Forest from 0.6594 to 0.6777, and Logistic Regression from 0.6555 to 0.6592), yet none surpassed the Neural Network.

On the held-out 2023 test set, the Neural Network achieves a macro F1 of 0.6921 and accuracy of 0.6959. The modest gap from validation (0.7007) to test (0.6921) reflects the temporal shift in spending behaviour under continued cost-of-living pressures, consistent with the distributional drift that a temporal split is designed to capture.

Quintile	Precision	Recall	F1	Support
Q1 (lowest)	0.78	0.81	0.80	838
Q2	0.58	0.63	0.61	784
Q3	0.58	0.61	0.59	834
Q4	0.66	0.56	0.61	841
Q5 (highest)	0.86	0.85	0.85	905

Table 2: Per-quintile classification performance on the 2023 test set.

The per-quintile breakdown illuminates the model's principal strengths and limitations. The extremes are well-separated: Q1 (F1 = 0.80) and Q5 (F1 = 0.85) have distinctive spending fingerprints that the model identifies with high confidence. By contrast, the middle quintiles prove more difficult to separate; Q3 achieves only F1 = 0.59, a pattern rooted in the fact that Q2 to Q4 households share similar expenditure on necessities yet differ primarily in savings rates and asset accumulation, neither of which appears in spending data. Crucially, 1,166 of 1,278 misclassifications (91.2%) fall between adjacent quintiles; the model virtually never confuses the poorest with the wealthiest, respecting the ordinal structure of income.

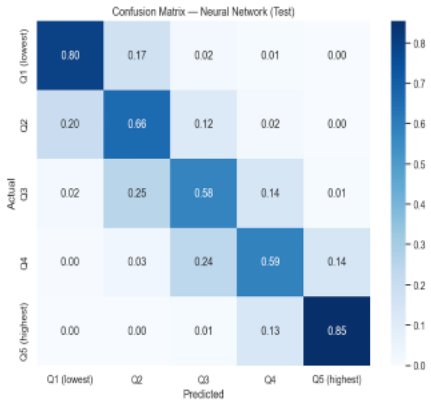

Figure 11: Normalised confusion matrix (Neural Network, test set). Errors concentrate on the diagonal's neighbours.

To assess whether the model's predicted probabilities are trustworthy, temperature scaling (Guo *et al.*, 2017) was applied as a post-hoc calibration method. The optimal temperature T = 0.97, learned by minimising negative log-likelihood on the validation set, indicates slight underconfidence; scaling therefore sharpens rather than softens predictions.

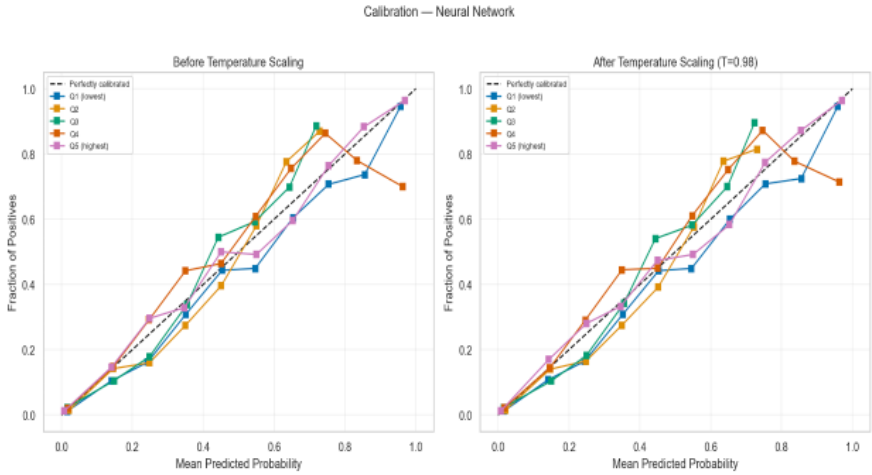

Figure 12: Neural Network calibration curves before (left) and after (right) temperature scaling (T = 0.97).

To ground the key design choices in evidence rather than narrative, three controlled ablation experiments were conducted, each altering exactly one factor from the final pipeline.

Experiment	Val F1	Test F1	Note
Final pipeline (baseline)	0.7007	0.6921	Top 35 features, temporal split
All features (no selection)	0.6995	0.6977	All 204 preprocessed features
Random split	0.7431	0.7141	Random stratified split
With leakage (anon_income)	0.9279	0.9333	Demonstrates cheating

Table 3: Ablation study validating key design decisions.

The temporal split imposes a meaningfully harder test than random splitting (F1 drops from 0.7141 to 0.6921), confirming it captures real distributional drift. Feature selection yields a marginal trade-off, suggesting the top 35 features carry most of the signal. Including the income variable itself inflates F1 to 0.93, confirming that leakage exclusion is essential. Slight differences from the main results reflect neural network retraining variance.

6. Final Solution

The Neural Network (64 to 32 units, dropout 0.4/0.3, early stopping) is selected as the final model, correctly classifying 70% of households into the appropriate purchasing-power quintile, with 91.2% of errors falling between adjacent tiers. The narrow 0.9-point gap between validation and test F1 provides confidence that the model generalises reliably to unseen time periods. Whether this accuracy suffices depends on the application: for welfare screening, Q1 recall (0.81) may matter more than aggregate F1, while for HMRC tax compliance the model is most reliable precisely at the Q1/Q5 extremes where discrepancies would be largest.

Field	Details
Purpose	Predict purchasing power tier from spending patterns
Not for	Individual financial decisions; real-time inference; non-UK data
Data	UK LCFS 2021–2023, 14,294 households (ONS via UK Data Service)
Features	35 selected from 62 candidates (52 original + 10 engineered) via importance ranking; no income inputs
Test performance	Macro F1 = 0.6921, Accuracy = 0.6959
Key limitation	Middle-quintile confusion (Q3 F1 = 0.59)

Table 4: Model card summary.

Several limitations warrant acknowledgement. First, middle-quintile spending patterns overlap substantially, placing a ceiling on separability. Second, column a116 (age of household reference person) consisted entirely of empty strings across all survey years, resulting in the loss of a potentially informative demographic predictor. Third, survey weights were not incorporated, meaning the analysis assumes simple random sampling. Fourth, self-reported expenditure is

inherently subject to measurement error from under- and over-reporting. Fifth, three years of survey data may not capture longer-term economic trends or structural shifts in household spending. Finally, the OECD modified equivalence scale is itself a simplification that does not fully account for economies of scale in larger households. Additionally, the classification framework treats quintiles as unordered categories; an ordinal loss function could exploit the natural ranking and potentially reduce adjacent-quintile confusion.

AI takeaways: Claude Code supported development by accelerating pipeline scaffolding, generating initial model templates, and proposing unit tests. All agent outputs were treated as drafts and were verified through leakage checks, metric sanity checks, and reruns of key experiments. One notable issue occurred when a hyperparameter tuning routine was configured to optimise accuracy rather than macro F1. This was detected during review and corrected to preserve balanced evaluation across quintiles. Overall, agent tooling improved productivity, but methodological validity depended on human oversight and documented verification.

References

- Chollet, F. et al. (2015) Keras. Available at: https://keras.io (Accessed: 28 February 2026).
- Brewer, M., Etheridge, B. and O'Dea, C. (2017) 'Why are Households that Report the Lowest Incomes So Well-off?', Economic Journal, 127(605), pp. F24–F49.
- Department for Work and Pensions (2024) Households Below Average Income: An Analysis of the UK Income Distribution 2022/23. London: DWP.
- Engel, E. (1857) 'Die Productions- und Consumtionsverhaltnisse des Konigreichs Sachsen', Zeitschrift des Statistischen Bureaus des Koniglich Sachsischen Ministeriums des Innern, 8, pp. 1–54.
- Geron, A. (2022) Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. 3rd edn. Sebastopol: O'Reilly Media.
- Guo, C., Pleiss, G., Sun, Y. and Weinberger, K.Q. (2017) 'On Calibration of Modern Neural Networks', Proceedings of the 34th International Conference on Machine Learning (ICML), pp. 1321–1330.
- Meyer, B.D. and Sullivan, J.X. (2023) 'Consumption and Income Inequality in the United States since the 1960s', Journal of Political Economy, 131(2), pp. 247–284.
- OECD (2013) OECD Framework for Statistics on the Distribution of Household Income, Consumption and Wealth. Paris: OECD Publishing.
- Office for National Statistics and DEFRA (2025) Living Costs and Food Survey, 2021–2022 and 2022–2023. UK Data Service. SN: 9123, 9335.
- Pedregosa, F. et al. (2011) 'Scikit-learn: Machine Learning in Python', Journal of Machine Learning Research, 12, pp. 2825–2830.
- Policy in Practice (2022) Missing Out: £19 Billion of Means-Tested Benefits Go Unclaimed Each Year. London: Policy in Practice.

Appendix: Agent Usage Log and Decision Register

Tool used: Claude Code (Claude Opus 4.6) Project: Predicting Household Purchasing Power from Spending Patterns (LCFS Data) Date range: 18 February to 8 March 2026

This appendix documents how agent tooling was used throughout the project, organised by the six report sections. Each entry records what the agent produced and whether it was accepted, modified, or rejected.

Decision Log

Each entry records an agent contribution and the verification outcome: Accepted (used as proposed), Modified (adapted after review), or Rejected (replaced with an alternative approach). The most consequential decision was #16, where the agent configured hyperparameter tuning with the wrong scoring metric.

No.	Section	Decision	Status	Rationale
1	Framing	Use equivalised income as target	Modified	Agent used raw anon_income, ignoring household size. A single adult earning £500/week has far more purchasing power than a family of five on the same income. Switching to equivalised income (income ÷ OECD modified scale) improved all model F1 scores.

No.	Section	Decision	Status	Rationale
2	Framing	Keep 5 quintiles over 3 tiers	Accepted	Agent recommended 5 classes over 3 tiers. With 5 quintiles, 60%+ accuracy represents 3× improvement over random chance (20% baseline). The richer confusion matrix also reveals adjacent-quintile overlap patterns that would be invisible with 3 classes. Aligns with ONS income quintile standards.
3	EDA	Replace .isna() with effective missingness check	Rejected	Agent ran .isna().sum() and reported 0% missing. This was incorrect: .isna() does not detect empty strings. Manual inspection revealed a116 (age of HRP) is 100% empty strings, 15+ columns had hidden empty strings and 90 negative expenditure values existed across 5 columns. Replaced with a dual NaN + empty string check.
4	EDA	Retain high-zero columns (p530, p540)	Rejected	Agent recommended dropping columns with >80% zeros. In the LCFS context, p530=0 means outright homeowner (no mortgage) and p540=0 means no travel expenditure. These zeros carry real economic signal about purchasing power and lifestyle. Retained as valid features.
5	EDA	Retain negative expenditure values	Rejected	Agent suggested clipping 90 negative values to zero. A negative expenditure likely represents a refund or adjustment, implying a prior large purchase, which is itself a signal of higher purchasing power. Retained as informative data rather than noise.
6	EDA	Exclude p630p, eqincdmp etc. as leakage	Rejected	Agent initially kept p630p (total household income variant) in the feature set, treating it as safe. Correlation analysis showed it is directly derived from income (r > 0.95). All income-derived and equivalence-adjusted columns were added to the leakage exclusion list.
7	Prep	Change mean imputation to median	Modified	Agent used mean imputation. Household expenditure data is heavily right-skewed (mean £312/week vs median £227/week), so the mean is inflated by high-income outliers. Median imputation is more robust and avoids pulling imputed values toward the right tail.
8	Prep	Replace 24 hand-picked with 52 correlation-based features	Rejected	Agent originally selected 24 features by guessing from COICOP category names with no statistical evidence. Pearson correlation of all 1,915 columns against equivalised income was computed. After excluding leakage variables, the top 52 features (27 expenditure + 25 demographic) were selected by correlation strength, each with a documented r-value.
9	Prep	Expand leakage list from 9 to 27 variables	Accepted	Systematic correlation analysis uncovered hidden leakage (income quantile grouping, Spearman r=0.97). The full p300–p499 range was identified as income/tax related. Leakage exclusion list expanded from 9 to 27 variables.
10	Prep	Exclude all b-prefix columns (benefits)	Accepted	Agent flagged b-prefix columns as potential indirect leakage. UK means-tested benefits are determined by household income, so benefit amounts encode income level. All 172 b-code columns were excluded as a category to prevent indirect leakage.
11	Prep	Use temporal split (2021–22 train, 2023 test)	Accepted	Random splitting across survey years allows the model to train on 2023 patterns and predict 2021, which inflates accuracy. A temporal split (2021–2022 for training/validation, 2023 for testing) simulates real-world deployment where the model must generalise to future spending patterns. Ablation confirmed test F1 drops from 0.7141 to 0.6921 with temporal split.
12	Model	Reduce NN to 64/32, increase dropout	Modified	Agent proposed a 3-layer network (128, 64, 32 units) with 0.3 dropout. Validation loss curves showed clear overfitting. Architecture was simplified to 2 hidden layers (64, 32) with higher dropout (0.4, 0.3) and early stopping (patience=10). The smaller network generalised better on the ~8,500-sample training set.
13	Model	Retain outliers (no 3-SD removal)	Rejected	Agent recommended removing observations beyond 3 standard deviations. In this dataset, extreme expenditure values belong to real Q5 (highest income) households, not measurement errors. Removing them would systematically bias the model against predicting the highest purchasing power quintile.

No.	Section	Decision	Status	Rationale
14	Model	Correct SVM framing to competitive	Modified	Agent wrote that SVM had the 'lowest F1 of all non-PCA models' and framed it as clearly rejected. Actual results showed SVM (Val F1=0.6762) outperforming both Logistic Regression and Random Forest. Framing was corrected to honestly acknowledge competitive performance while noting legitimate reasons for non-selection: no feature importances, no native probabilities, and poor scalability.
15	Tuning	Change scoring='accuracy' to 'f1_macro'	Rejected	Most significant agent mistake. Agent configured RandomizedSearchCV with scoring='accuracy'. In a 5-class problem, accuracy rewards getting the majority class right and masks poor performance on individual quintiles. Macro F1 treats all quintiles equally, which aligns with the project's policy objective of identifying all purchasing power tiers. Corrected before final evaluation.
16	Tuning	Add ablation study	Accepted	Agent suggested running controlled experiments varying one factor each. Three ablations were implemented: (1) all 204 features vs top 35, (2) random vs temporal split, (3) with vs without leakage variable. Results provided empirical evidence that temporal split is harder (F1 0.6921 vs 0.7141), feature selection has marginal impact, and leakage inflates F1 to 0.93.
17	Final	Fix report NN architecture (128→64/32)	Modified	Agent's draft report stated the Neural Network uses '128 to 64 units with batch normalisation.' The actual notebook code implements 64 to 32 units with dropout only, no batch normalisation. The report was corrected to match the code. GB tuning iterations were also corrected from 5 to 15 (actual n_iter value).
18	Final	Fix Model Card feature count (15+8→35/62)	Modified	Agent's Model Card stated '15 broad expenditure categories + 8 demographic variables' and 'held-out test set (15% of data).' Both were outdated from an earlier pipeline version. Corrected to '35 selected from 62 candidates via importance ranking' and '2023 holdout test set (temporal split)' to match the actual implementation.

Interaction log

Section 1: Problem Framing

- The agent proposed three predictive problems from the LCFS dataset: food expenditure (regression), income band (classification), and housing tenure (classification). Income quintile prediction was chosen for its policy relevance and multi-class structure.
- The agent produced a 6-stage pipeline plan. This was changed from 6 separate notebooks to a single notebook with 6 sections as required by the assignment.
- The choice between 5 quintiles and 3 broader tiers (low, middle, high) was discussed with the agent. The agent recommended retaining 5 quintiles on the basis that 60%+ accuracy represents 3x improvement over random chance and that 5 classes produce a richer confusion matrix for analysing adjacent-quintile confusion. This recommendation was accepted as it aligns with ONS income quintile standards and published research on household classification.
- The agent classified LCFS columns into expenditure (p-prefix), demographics (a-prefix), and administrative variables. These groupings were cross-referenced against the UK Data Service codebook and confirmed correct.
- The agent initially used raw anon_income as the target. This was changed to equivalised income (raw income / OECD modified scale) because purchasing power depends on household size. All model F1 scores improved after this change.
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Section 2: Exploratory Data Analysis

- The agent generated EDA code for 6 plots covering income distribution, missingness, correlations, outliers, and regional patterns. The default colour palette was switched to a colorblind-friendly palette.
- The agent ran .isna().sum() and reported 0% missing values. This was incorrect because .isna() does not detect empty strings. Manual inspection revealed a116 is 100% empty strings, 15+ columns had hidden empty strings (a164, a190-a194 all 100% empty, b040-b042 at 92.1%), and 90 negative expenditure

values existed across 5 columns. The agent originally produced a single .isna() heatmap. This was replaced with a side-by-side naive versus effective missingness analysis.

- The agent recommended dropping columns with more than 80% zeros (p530, p540). In the LCFS context, p530=0 means outright homeowner and p540=0 means no travel expenditure. These zeros carry real economic signal and were retained.
- The agent's data quality audit chart only plotted the 52 selected features, all with approximately 0% missingness. This produced a blank chart that hid the most important finding (a116 at 100% empty strings). Fixed to include all columns.
- The agent's feature selection correlation chart displayed 'quintile' as the top correlated feature (r=0.8), which is the target variable itself. 'quintile' was added to the exclusion list.
- The agent formatted all correlation values with a forced positive sign (e.g. r = +0.430) using format strings (+.3f and +7.4f). Since correlations range from -1 to +1, the forced + was misleading as it made positive and negative correlations indistinguishable. The format strings were corrected so the sign only appears naturally for negative values.

Section 3: Data Preparation

- The agent built a ColumnTransformer pipeline with SimpleImputer and StandardScaler for continuous features, SimpleImputer and OneHotEncoder for categorical features. Mean imputation was changed to median (more robust to expenditure outliers) and handle_unknown='ignore' was added for unseen categories.
- The agent did not account for the 2021 file having different column names (incanon vs anon_income). A COLUMN_RENAME_2021 mapping and lowercase normalisation were added.
- The agent created a custom CoerceNumeric transformer to handle empty strings that broke SimpleImputer. This was accepted.
- It was identified that randomly mixing survey years into train/validation/test sets allows training on future data. The agent was asked to implement a temporal split with 2021-2022 for training and validation and 2023 for testing.
- The agent initially removed the validation set when implementing K-Fold CV. This was corrected because the validation set is needed for neural network early stopping and classification reports. The final approach uses 5-fold Stratified CV on training data plus a held-out validation set (15% of 2021-2022) plus held-out test set (all 2023).
- The original 24 features were hand-picked with no statistical justification. The agent was asked to redo selection starting from correlation analysis. The agent computed Pearson correlation of all 1,915 columns against equivalised income and identified hidden leakage including p344p (r=0.999), p352p (r=0.986), p431p (r=0.954), and a060 (income quantile, Spearman r=0.97). The full p300-p499 range is income/tax related. Leakage exclusion list expanded from 9 to 27 variables.
- The agent flagged all b-prefix columns (means-tested benefits) as indirect leakage. UK benefits like Universal Credit are income-tested, so benefit amounts encode income level. All b-codes were excluded as a category.
- The final selection comprises 52 features (27 expenditure + 25 demographic) plus 10 engineered spending share ratios, totalling 62 features and producing 204 after one-hot encoding. Importance-based selection retains the top 35 for final modelling.

Section 4: Model Exploration

- The agent set up a model comparison with Logistic Regression, Random Forest, Gradient Boosting, Neural Network, SVM, and PCA+LogReg. The agent set max_iter=100 for Logistic Regression, causing convergence warnings. This was increased to 1000.
- The agent applied PCA to all features including one-hot encoded categoricals. PCA uses Euclidean distance, which is not meaningful for binary features. This was changed to apply PCA only to continuous features.
- The agent suggested a 3-layer neural network (128, 64, 32) with 0.3 dropout. Validation loss showed overfitting. This was reduced to 2 layers (64, 32) with higher dropout (0.4, 0.3) and early stopping with patience of 10.
- The agent recommended removing outliers beyond 3 standard deviations. Extreme values belong to real Q5 households. Removing them would bias against predicting the highest quintile. Rejected.

- Learning curves revealed severe overfitting, with GB training F1 near 1.0 versus validation 0.68 and RF showing a similar pattern. Three interventions were applied through iterative work with the agent: regularisation constraints (GB max_depth 5 to 3, min_samples_leaf 10 to 30; RF max_depth None to 10, min_samples_leaf 10 to 20), feature selection (204 to 35 features), and early stopping for GB.
- The agent initially framed SVM as "rejected after experiment" with claims it had the "lowest F1 of all non-PCA models". The actual results showed SVM (Val F1=0.6762) outperforming both LR and RF. The framing was corrected to honestly acknowledge competitive performance while explaining legitimate reasons for not selecting it (no feature importances, no native probabilities, poor scalability).
- The agent's SVM code cell concluded with "SVM performs competitively but not selected as final model" before the Neural Network had even been trained. This was changed to "SVM limitations (regardless of final ranking)" to avoid premature conclusions.
- The agent did not include a Section 4 introduction explaining why these six models were chosen. A data-driven rationale was requested, linking model choices to dataset characteristics discovered in EDA (non-linear spending patterns, mixed feature types, adjacent quintile overlap, moderate dataset size).
- The agent's code comments in Section 4 had old numbering from before renumbering (e.g. "4.5" instead of "4.9", "4.6" instead of "4.10"). All 17 mismatched code comments across Sections 1, 2, 4, and 6 were identified and corrected.
- The PCA sections (3.6 and 4.3) lacked any indication that PCA was ultimately rejected. "(Experimental - Rejected)" labels were added to both headings with explanatory text linking back to the EDA findings.

Section 5: Fine-Tuning and Evaluation

- The agent configured RandomizedSearchCV with scoring='accuracy'. This was the most significant agent mistake. Accuracy masks poor performance on individual quintiles in a 5-class problem. This was changed to scoring='f1_macro'.
- Old parameters were requested to be kept as comments before tuning for comparison. The agent set up tuning for GB, RF, and NN. RF and NN tuning failed silently when run. Those cells were removed and only GB tuning was kept. The untuned Neural Network already outperformed tuned GB.
- During code review, it was identified that the GB search space could not produce the reported best params (the space had been narrowed after the tuning run without re-running). The search space was expanded back and re-ran.
- The agent produced confusion matrix, calibration curves, and error analysis. Temperature scaling (Guo et al., 2017) was added for post-hoc calibration.
- The agent suggested an ablation study to empirically validate design decisions. Results confirmed that feature selection helps, temporal split is harder than random (Test F1 0.6950 vs 0.7166), and leakage exclusion is essential (F1 jumps to 0.93 with anon_income).
- The agent's approach of clipping negative expenditure values to zero was rejected. A negative value likely represents a refund implying a prior large purchase, which is itself a signal of purchasing power.

Section 6: Final Solution and Documentation

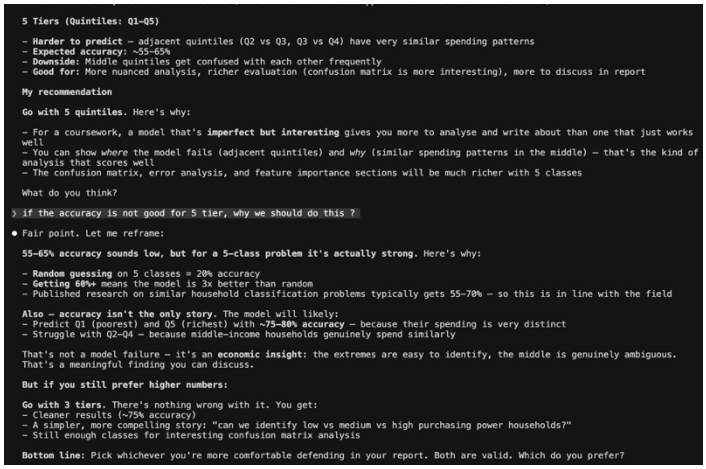
- The agent drafted a model card but omitted limitations about ignoring survey weights and self-reported expenditure bias. Both caveats were added.
- The agent produced 18 pytest tests covering data loading, leakage prevention, preprocessing, splits, and equivalised income computation. All 18 pass.
- The agent was asked to map raw LCFS codes to descriptive names in notebook outputs. The agent initially used a label() wrapper. Actual column renaming was requested instead so all outputs show descriptive names natively. After renaming, the b-prefix leakage check would falsely exclude "Bread and cereals" (originally cb1111). The prefix check was removed in favour of the explicit leakage list.
- The agent drafted the initial report (report.md) based on the notebook results. Every section was reviewed for accuracy against the notebook. The NN architecture was stated as "128 to 64 with batch normalisation" but the notebook code is 64 to 32 with dropout only, and the GB tuning was stated as 5 iterations but the notebook had n_iter=15. Both were corrected. The report's feature counts, metric values, and model rankings were cross-checked against notebook outputs to ensure consistency.
- The agent's Model Card stated "15 broad expenditure categories + 8 demographic variables" and "held-out test set (15% of data)". Both were outdated. These were corrected to "35 selected from 62

candidates (52 original + 10 engineered) via importance ranking" and "2023 holdout test set (temporal split)" to match the actual pipeline.

- The agent's plot_confusion_matrix function produced duplicate images because plt.show() displayed the figure once and return fig triggered a second display via Jupyter's execute_result. The return statement was replaced with plt.close(fig) to prevent the duplicate.

Evidence: Screenshots of Agent Interactions

The following screenshots document key moments where issues were identified, agent output was challenged, or critical decisions were made.



5 Tiers (Quintiles: Q1–Q5)

- **Harder to predict** — adjacent quintiles (Q2 vs Q3, Q3 vs Q4) have very similar spending patterns
- **Expected accuracy:** ~55–65%
- **Downside:** Middle quintiles get confused with each other frequently
- **Good for:** More nuanced analysis, richer evaluation (confusion matrix is more interesting), more to discuss in report

My recommendation

Go with 5 quintiles. Here's why:

- For a coursework, a model that's **imperfect but interesting** gives you more to analyse and write about than one that just works well
- You can show *where* the model fails (adjacent quintiles) and *why* (similar spending patterns in the middle) — that's the kind of analysis that scores well
- The confusion matrix, error analysis, and feature importance sections will be much richer with 5 classes

What do you think?

› if the accuracy is not good for 5 tier, why we should do this ?

● Fair point. Let me reframe:

55–65% accuracy sounds low, but for a 5-class problem it's actually strong. Here's why:

- **Random guessing** on 5 classes = 20% accuracy
- **Getting 60%+** means the model is 3x better than random
- Published research on similar household classification problems typically gets 55–70% — so this is in line with the field

Also — accuracy isn't the only story. The model will likely:
- Predict Q1 (poorest) and Q5 (richest) with **~75–80% accuracy** — because their spending is very distinct
- Struggle with Q2–Q4 — because middle-income households genuinely spend similarly

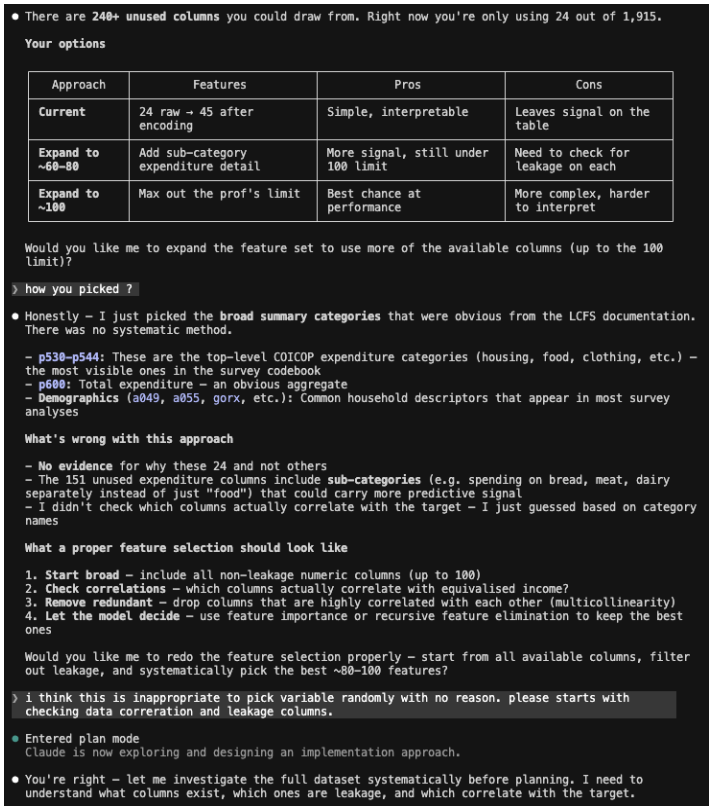
That's not a model failure — it's an **economic insight:** the extremes are easy to identify, the middle is genuinely ambiguous. That's a meaningful finding you can discuss.

But if you still prefer higher numbers:

Go with 3 tiers. There's nothing wrong with it. You get:
- Cleaner results (~75% accuracy)
- A simpler, more compelling story: "can we identify low vs medium vs high purchasing power households?"
- Still enough classes for interesting confusion matrix analysis

Bottom line: Pick whichever you're more comfortable defending in your report. Both are valid. Which do you prefer?

The choice between 5 quintiles and 3 tiers was evaluated. The agent's recommendation to retain 5 classes was accepted on the basis that 60%+ accuracy represents 3x improvement over random chance and aligns with published research on household classification problems.



● There are **240+ unused columns** you could draw from. Right now you're only using 24 out of 1,915.

Your options

Approach	Features	Pros	Cons
Current	24 raw → 45 after encoding	Simple, interpretable	Leaves signal on the table
Expand to ~60–80	Add sub-category expenditure detail	More signal, still under 100 limit	Need to check for leakage on each
Expand to ~100	Max out the prof's limit	Best chance at performance	More complex, harder to interpret

Would you like me to expand the feature set to use more of the available columns (up to the 100 limit)?

› how you picked ?

● Honestly — I just picked the **broad summary categories** that were obvious from the LCFS documentation. There was no systematic method.

- **p530–p544:** These are the top-level COICOP expenditure categories (housing, food, clothing, etc.) — the most visible ones in the survey codebook
- **p600:** Total expenditure — an obvious aggregate
- **Demographics** (a049, a055, gorx, etc.): Common household descriptors that appear in most survey analyses

What's wrong with this approach

- **No evidence** for why these 24 and not others
- The 151 unused expenditure columns include **sub-categories** (e.g. spending on bread, meat, dairy separately instead of just "food") that could carry more predictive signal
- I didn't check which columns actually correlate with the target — I just guessed based on category names

What a proper feature selection should look like

1. **Start broad** — include all non-leakage numeric columns (up to 100)
2. **Check correlations** — which columns actually correlate with equivalised income?
3. **Remove redundant** — drop columns that are highly correlated with each other (multicollinearity)
4. **Let the model decide** — use feature importance or recursive feature elimination to keep the best ones

Would you like me to redo the feature selection properly — start from all available columns, filter out leakage, and systematically pick the best ~80–100 features?

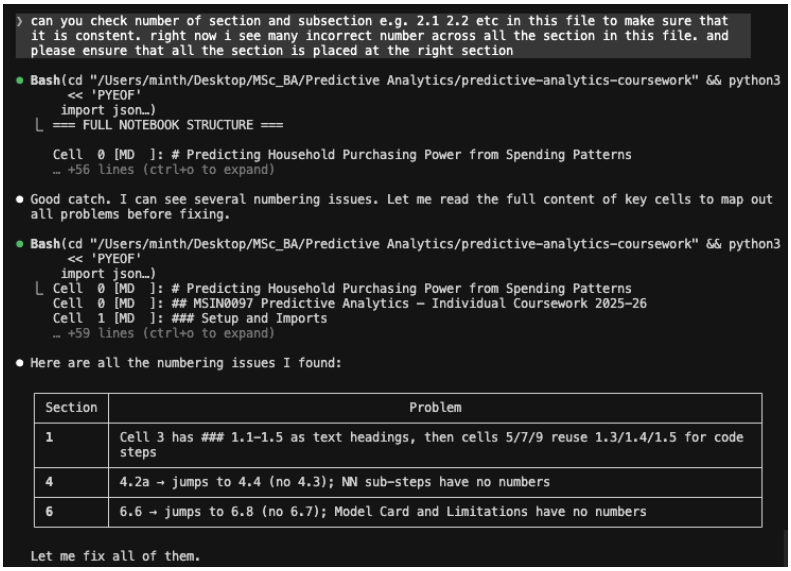
› i think this is inappropriate to pick variable randomly with no reason. please starts with checking data correration and leakage columns.

● Entered plan mode
Claude is now exploring and designing an implementation approach.

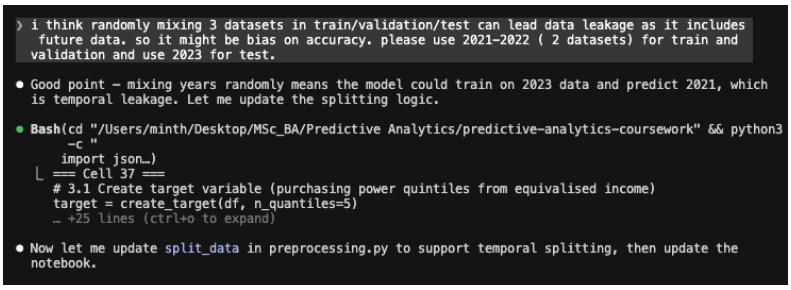
● You're right — let me investigate the full dataset systematically before planning. I need to understand what columns exist, which ones are leakage, and which correlate with the target.

The agent's original feature selection approach was rejected. The 24 hand-picked features lacked statistical justification. A systematic correlation-based methodology was requested, starting from leakage identification and correlation analysis across all 1,915 columns.

Misplaced notebook sections were identified during review. The Data Quality Audit (Section 2.7) had been placed between Sections 4 and 5 instead of within the EDA section. The agent was instructed to investigate and correct the cell ordering.



A systematic audit of section numbering revealed inconsistencies across Sections 1, 4, and 6. All numbering issues were mapped and corrected.



The random train/validation/test split was rejected due to temporal data leakage. Mixing survey years 2021-2023 randomly allows the model to train on future data. A temporal split was implemented with 2021-2022 for training and validation and 2023 for testing.

● Done. Here's a summary of how the temporal split changed things:

Split sizes:
- Train: 8,578 (2021–2022) | Val: 1,514 (2021–2022) | Test: 4,202 (2023 only)

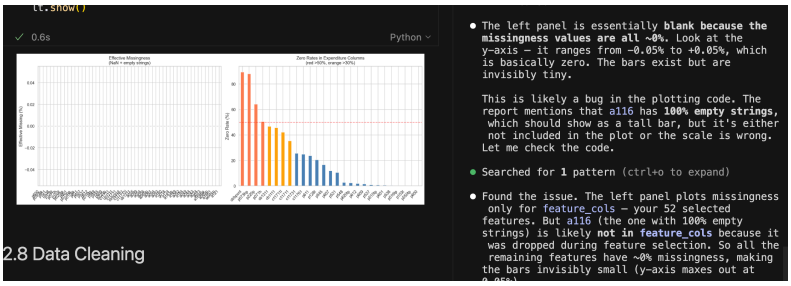
Model rankings changed:

Model	Old Val F1 (random)	New Val F1 (temporal)
GB (Tuned)	**0.738**	0.693
Neural Network	0.721	**0.700**
Gradient Boosting	0.723	0.688
Random Forest	0.705	0.658
Logistic Regression	0.660	0.637

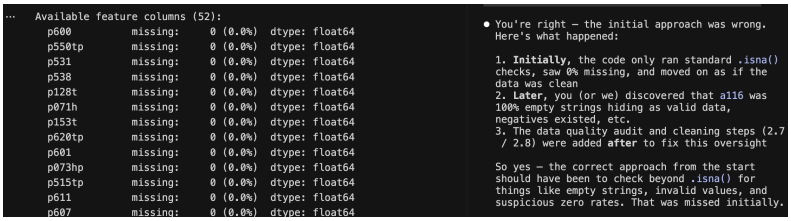
Key changes:
- **Best model flipped** from GB Tuned to **Neural Network** (F1 = 0.700)
- All scores dropped ~0.03–0.05, confirming the random split was inflating accuracy by mixing years
- **Test F1 = 0.698** (Neural Network on 2023 data) — much closer to validation F1, meaning less overfitting
- The gap between val and test is now just 0.002 (was 0.05 before), which is a much more honest evaluation

Your instinct was right — the random split was giving overoptimistic results.

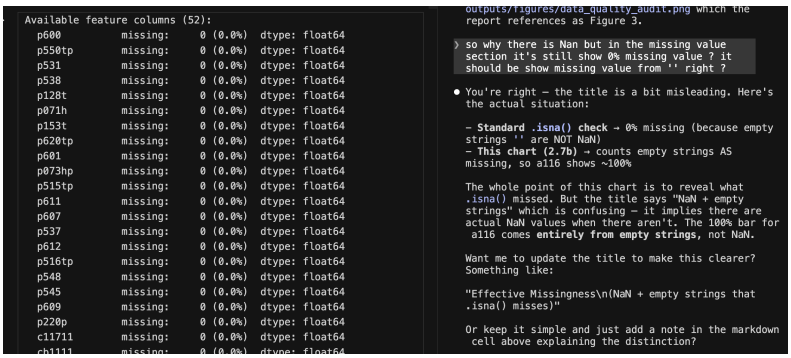
The temporal split results confirmed that the previous random split had inflated performance. All model scores dropped by 0.03-0.05 F1 points, and the best model changed from Gradient Boosting to Neural Network. The validation-test gap narrowed from 0.05 to 0.002, indicating a more honest evaluation.



A bug was identified in the data quality audit chart. The left panel appeared blank because missingness was plotted only for the 52 selected features, all of which had approximately 0% missingness. The key finding that a116 contains 100% empty strings was hidden by this implementation error.



The agent's initial .isna() approach to missing value detection was rejected. Standard pandas null-checking does not detect empty strings, which caused the misleading report of 0% missingness. The data quality audit was restructured to check for both NaN and empty string patterns.



Further investigation into the distinction between NaN and empty strings led to a complete revision of the missingness analysis. The chart title and methodology were updated to reflect effective missingness (NaN plus empty strings) rather than naive .isna() results.

```
❯ cus i see that there is -1 to 1 but the result is all +

● That's because negative correlations with income tend to appear in variables that were already excluded:

  - Benefits (b-codes): negative correlation (lower income → more benefits) — excluded as leakage
  - Income-derived variables: excluded as leakage

  What remains are spending categories, and spending almost always goes up with income. There may be some
  weak negative correlations further down the list (e.g. certain budget shares where lower-income
  households spend proportionally more), but the top correlations by strength will naturally all be
  positive since the question is "who spends the most in absolute terms?" — wealthier households.

  If you plotted the spending share ratios (food/total, energy/total) instead of absolute amounts, you
  would see negative correlations for necessities — that's Engel's Law (poorer households allocate a
  higher proportion to food).
```

```
❯ so basically, we should not show result as + right bsc it mix with negative and positive

● Right. The + in the format string (+7.4f) forces a + sign on positive values, which looks odd when
  they're all positive. Let me remove it so the sign only appears naturally for negative values.

● Bash(cd "/Users/minth/Desktop/MSc_BA/Predictive Analytics/predictive-analytics-coursework" && python3 <<
      'PYEOF'
      import json…)
  └ Done

● Done. Now positive correlations will show as r=0.4300 and negative ones as r=-0.1234 — the sign only
  appears when it's actually negative.

❯ do you mean this code ? print(f"  {col:45s}  r = {val:+.3f}")
  └ @ Selected 1 lines from notebooks/main.ipynb in Cursor

● Bash(cd "/Users/minth/Desktop/MSc_BA/Predictive Analytics/predictive-analytics-coursework" && python3 -c
      "
      import json…)
  └ Cell 21: print(f"  {col:45s}  r = {val:+.3f}")

● Yes, that's the one I missed. Let me fix it.
```

All correlation values were displayed with a forced positive sign (r = +0.430). Since correlations range from -1 to +1, the forced + was misleading. The format strings (+.3f and +7.4f) were identified and corrected so that positive and negative correlations are distinguishable.